

How Reliable are LLMs as Knowledge Bases? Re-thinking Factuality and Consistency

Danna Zheng¹, Mirella Lapata¹, Jeff Z. Pan^{1,2}

¹ School of Informatics, University of Edinburgh, UK

² Huawei Edinburgh Research Centre, CSI, UK

dzheng@ed.ac.uk, mlap@inf.ed.ac.uk, <http://knowledge-representation.org/j.z.pan/>

Abstract

Large Language Models (LLMs) are increasingly explored as knowledge bases (KBs), yet current evaluation methods focus too narrowly on knowledge retention, overlooking other crucial criteria for reliable performance. In this work, we rethink the requirements for evaluating reliable LLM-as-KB usage and highlight two essential factors: factuality, ensuring accurate responses to seen *and* unseen knowledge¹, and consistency, maintaining stable answers to questions about the same knowledge. We introduce UnseenQA, a dataset designed to assess LLM performance on unseen knowledge, and propose new criteria and metrics to quantify factuality and consistency, leading to a final reliability score. Our experiments on 26 LLMs reveal several challenges regarding their use as KBs, underscoring the need for more principled and comprehensive evaluation.

1 Introduction

Large Language Models (LLMs), pretrained on vast text corpora, have demonstrated significant capabilities in encoding knowledge without explicit supervision. The continuous release of new LLMs, evaluated on benchmarks like TriviaQA (Joshi et al., 2017) and Natural Questions (Kwiatkowski et al., 2019), highlights their improving ability to answer fact-based queries. This progress has fueled interest in employing LLMs as knowledge bases (KBs) for various applications and developing techniques to edit model knowledge (Wang et al., 2024c,b,a) or mitigate hallucinations (Zhang et al., 2024b,a; Yu et al., 2024).

However, a critical question remains underexplored: *What criteria should an LLM meet to function reliably as a KB?* Current research often assumes that knowledge retention alone is sufficient (Sun et al., 2023; Wang et al., 2021; Roberts

et al., 2020). Existing evaluations generally follow two approaches: (1) converting knowledge graphs into natural language questions and assessing how many questions the LLM answers correctly (Petroni et al., 2019; Sun et al., 2023); and (2) pretraining LLMs on knowledge-rich text and measuring their accuracy on related questions (Wang et al., 2021; He et al., 2024).

These methods demonstrate that LLMs can recall information, but knowledge volume alone does not guarantee reliable performance as a KB. Beyond retention, it is essential to examine how LLMs handle factual queries—specifically, whether they respond accurately to seen knowledge and avoid making claims about unseen knowledge (*factuality*), and whether they provide consistent answers to questions about the same facts (*consistency*).

Factuality refers to the quality of being factual or based on fact. KBs hosted on servers or cloud platforms, offer precise answers or null responses when data is unavailable. In contrast, LLMs rely on probabilistic next-token prediction, which can lead to plausible but incorrect answers. As a result, LLM responses are typically correct, uninformative, or wrong. Existing methods for evaluating factuality often focus on the rate of correct answers in factual QA datasets (Chen et al., 2023; Wang et al., 2024d). However, as many studies (Lin et al., 2022; Sun et al., 2023) fail to specify whether the dataset’s knowledge was included in the LLM’s pre-training data, it is not possible to establish whether the model is genuinely factual. Secondly, it is misleading to equate a higher correct rate with greater factuality. As illustrated in Figure 1(a), a model with a higher correct rate might still be less factual if it produces a higher rate of wrong answers compared to a model with fewer errors overall.

Consistency refers to the quality of always behaving in the same way or having the same opinions. Traditional KBs achieve consistency through algorithms (Andersen and Pretolani, 2001) that de-

¹Seen knowledge refers to knowledge learned during training. Unseen knowledge is neither present in the model’s training data nor can be inferred from seen knowledge.

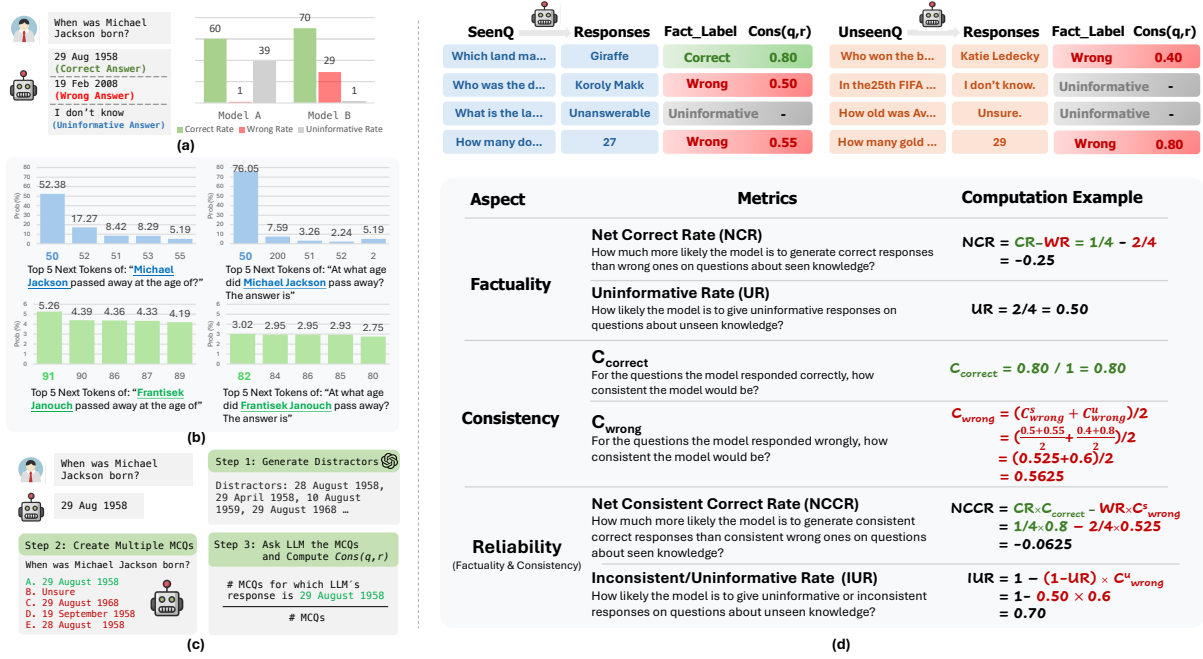


Figure 1: (a) An example illustrating three answer types: correct, wrong, and uninformative. Focusing only on the correct rate incorrectly suggests that Model B is better, even though Model A is more reliable with a similar correct rate and a much lower wrong rate. (b) Illustration of LLM inconsistency with DAVINCI-002 (temperature is set to 0). Questions in the top focus on seen knowledge, with probability distribution mass concentrated on one prediction. Questions in the bottom focus on unseen knowledge, where the distribution is more even. Drawing from such a distribution inevitably leads to inconsistencies. (c) Example computation for consistency score $Cons(q, r)$. The LLM’s original answer is shown in green, while distractors are red. (d) An example illustrating how to evaluate LLM-as-KB.

tect and resolve conflicts. In contrast, LLMs frequently exhibit inconsistent behavior (Elazar et al., 2021; Wang et al., 2022). Current research (Elazar et al., 2021; Jang et al., 2022; Hagström et al., 2023) evaluates LLM consistency using benchmarks involving paraphrasing, negation, or multilingual variations, favoring models that maintain consistent responses across diverse samples. However, we argue that expecting LLMs to always be consistent in fact-based responses is overly rigid. Unlike KBs, which store information in fixed locations, LLMs operate probabilistically. When the context has been learned during training, the probability distribution for predictions is concentrated; otherwise, it remains more uniform. Sampling from a uniform distribution naturally leads to inconsistencies. As shown in Figure 1(b), even with greedy decoding, slight distribution biases can cause variations in the top-selected words.

Given these issues, this paper seeks to define the criteria for a reliable LLM-as-KB when handling factual queries. In evaluating factuality, we consider both seen knowledge (contained within training data) and unseen knowledge, and consider

the negative effects of wrong answers. To assess performance on unseen knowledge, we introduce UnseenQA, a new dataset containing knowledge unavailable to LLMs trained before April 13, 2024. For consistency, we classify the correctness of responses and propose a novel method to compute the probability that an LLM can consistently provide the same response r to a question q .

We evaluate 26 popular LLMs and find that: 1) GPT-3.5-TURBO achieves balanced performance in both factuality and consistency. LLMs like LLAMA3-70B may achieve a higher correct rate on seen knowledge but exhibit higher wrong rates and consistency on wrong answers. 2) There is a correlation between factuality and consistency: more factual LLMs tend to be consistent in their responses, whether correct or wrong. 3) Larger LLMs perform worse on unseen knowledge and are more consistent even when providing wrong answers. 4) Fine-tuning techniques can improve performance on unseen knowledge. However, this often comes at the expense of performance on seen knowledge. 5) In-context learning (ICL) does not improve performance on seen knowledge, as it as

it generally increases or decreases both correct and wrong rates simultaneously. 6) Base LLMs tend to overestimate their knowledge on numerical and temporal questions.

We hope our work will draw the community’s attention to the multifaceted challenges of using LLMs-as-KBs, ultimately inspiring further research and innovation toward more reliable, robust, and principled methodologies.

2 What is a Reliable LLM-as-KB?

In simple terms, an LLM is a reliable KB if it consistently provides factual responses. As illustrating in Figure 1 (d), evaluating the reliability of LLMs as KBs primarily involves assessing two critical dimensions, namely factuality and consistency.

2.1 Factuality

We propose the following criteria for determining the factuality of LLMs-as-KBs:

Criterion 1.1: For seen knowledge, a factual LLM should demonstrate a high correct rate and a low wrong rate.

Criterion 1.2: For unseen knowledge, a factual LLM should demonstrate a high uninformative rate.

We next proceed to define evaluation metrics that operationalize these criteria. Let M denote an LLM. Let D_{seen} denote a QA dataset containing N open-ended factoid questions pertaining to knowledge the LLM ought to have seen during training. Let D_{unseen} denote a QA dataset with L open-ended factoid questions covering unseen knowledge. We further assume the LLM’s response to D_{seen} will be correct, uninformative, or wrong, while its response to D_{unseen} will be either uninformative or wrong.

METRIC 1.1: Net Correct Rate (NCR) measures how much more likely the model is to provide correct responses instead of wrong ones on D_{seen} questions. It is defined as:

$$\text{NCR} = \text{CR} - \text{WR} \quad (1)$$

$$\text{CR} = \frac{N_{\text{correct}}}{N} \quad \text{WR} = \frac{N_{\text{wrong}}}{N} \quad (2)$$

where N_{correct} and N_{wrong} are counts of correct and wrong responses, respectively.

NCR values range from -1 to 1 . A negative NCR suggests the model tends to provide misleading responses, while a positive NCR suggests a preference for correct responses. Consider again two models, A and B. According to Criterion 1.1, if model A has a higher correct rate and lower wrong rate compared to model B, then model A is better. Formally, if $\text{CR}_A - \text{CR}_B > \text{WR}_A - \text{WR}_B$, then model A is better than B. Algebraically, this is equivalent to $\text{CR}_A - \text{WR}_A > \text{CR}_B - \text{WR}_B$, i.e., $\text{NCR}_A > \text{NCR}_B$. Therefore, a higher NCR indicates a more factual model on seen knowledge.

METRIC 1.2: Uninformative Rate (UR) assesses whether the model is likely to provide uninformative responses to D_{unseen} questions. It is formulated as:

$$\text{UR} = \frac{L_{\text{uninformative}}}{L} \quad (3)$$

where $L_{\text{uninformative}}$ denotes the count of uninformative responses. UR ranges from 0 to 1. A higher UR indicates that the model is more likely to refrain from giving wrong responses when faced with unseen knowledge.

2.2 Consistency

We propose the following consistency criteria:

Criterion 2.1: The model is expected to be consistent in correct responses.

Criterion 2.2: The model is expected to be inconsistent in wrong responses.

We next define evaluation metrics corresponding to the criteria above. Let q refer to a question in either D_{seen} or D_{unseen} , and r denote model M ’s response to q . Inspired by Zheng et al. (2024), we measure consistency based on multiple-choice questions (MCQs). As shown in Figure 1 (c), we employ GPT-3.5-TURBO-INSRUCT to generate a set of distractor options similar to response r , and then create a group of MCQs. The consistency score for data point (q, r) is calculated as:

$$\text{Cons}(q, r) = \frac{\sum_{i=1}^{X_{\text{MCQs}}} [R_i = r]}{X_{\text{MCQs}}} \quad (4)$$

where X_{MCQs} is the total number of MCQs, R_i is model M ’s response for the i -th MCQ, and $[R_i = r]$ yields 1 when the model’s response R_i matches its original response r , and 0 otherwise. The consistency score $\text{Cons}(q, r)$ ranges from 0 to 1.

METRIC 2.1: $C_{correct}$ measures a model’s consistency in its correct responses and is defined as:

$$C_{correct} = \frac{\sum_{j=1}^{N_{correct}} Cons(q_j^{(c)}, r_j^{(c)})}{N_{correct}} \quad (5)$$

where $r^{(c)}$ refers to the response labeled as correct, and $q^{(c)}$ is the corresponding question. $C_{correct}$ ranges from 0 to 1. Based on Criterion 2.1, a higher $C_{correct}$ is desirable.

METRIC 2.2: C_{wrong} measures the consistency of an LLM when it provides wrong responses and is defined as:

$$C_{wrong} = \frac{C_{wrong}^s + C_{wrong}^u}{2} \quad (6)$$

where C_{wrong}^s/C_{wrong}^u refer to the consistency of an LLM when it provides wrong responses to questions about seen/unseen knowledge:

$$C_{wrong}^s = \frac{\sum_{j=1}^{N_{wrong}} Cons(q_j^{(w)}, r_j^{(w)})}{N_{wrong}} \quad (7)$$

$$C_{wrong}^u = \frac{\sum_{j=1}^{L_{wrong}} Cons(q_j^{(w)}, r_j^{(w)})}{L_{wrong}} \quad (8)$$

where $r^{(w)}$ and $q^{(w)}$ denote wrong responses and their corresponding questions. N_{wrong} and L_{wrong} are the counts of wrong answers on D_{seen} and D_{unseen} , respectively. C_{wrong} ranges from 0 to 1, and per Criterion 2.2, lower C_{wrong} is better.

2.3 Reliability (Factuality and Consistency)

Based on the criteria defined above, an LLM is reliable as a KB if it meets the following criteria when evaluated against factuality *and* consistency:

Criterion 3.1: For seen knowledge, an LLM should have a high rate of consistently correct responses and a low rate of consistently wrong responses.

Criterion 3.2: For unseen knowledge, a LLM should have a high rate of uninformative or inconsistent responses.

We next quantify these criteria are with the following two metrics.

METRIC 3.1: Net Consistently Correct Rate (NCCR) quantifies the model’s tendency to provide consistently correct responses compared to

consistently wrong ones for questions about seen knowledge. It is defined as:

$$NCCR = CCR - CWR \quad (9)$$

$$CCR = CR \times C_{correct}$$

$$CWR = WR \times C_{wrong}^s$$

NCCR ranges from -1 to 1 . NCCR values closer to 1 indicate an LLM is more reliable on seen knowledge. A negative NCCR suggests the model provides consistently wrong responses, while a positive NCCR suggests a preference for consistently correct responses.

METRIC 3.2: Inconsistent/Uninformative Rate (IUR) assesses whether an LLM is likely to provide uninformative or inconsistent wrong responses for questions about unseen knowledge. It is defined as:

$$IUR = 1 - (1 - UR)C_{wrong}^u \quad (10)$$

IUR ranges from 0 to 1 . A higher IUR value indicates the LLM functions as a more reliable KB on unseen knowledge.

3 Experimental Setup

3.1 LLM Selection

We evaluate 26 popular LLMs, including [GPT-3.5-TURBO](#), [FLAN-T5](#), [LLAMA1](#), [LLAMA2](#), [LLAMA3](#), [MISTRAL](#), [GEMMA](#), and [PHI2](#). Detailed descriptions of the evaluated LLMs are provided in Table 4 in Appendix B. Detailed descriptions are available in Table 4 in Appendix B. We test LLMs of different sizes: small (0.08B–3B), medium (7B–13B), and large (65B–70B). "Fine-tuned LLMs" refer to those fine-tuned via instruction-tuning or reinforcement learning from human feedback (e.g., LLAMA3INSTRUCT-8B), while "base LLMs" refer to models without fine-tuning (e.g., LLAMA3-8B).

3.2 Datasets

SeenQA SeenQA is a composite dataset comprising 3,000 questions sourced from the test sets (or development sets, where test sets were unavailable) of [Natural Questions](#), [TriviaQA](#), and [PopQA](#) (see Appendix A). All three datasets are derived from Wikipedia, with a knowledge cutoff date no later than December 2018. Given that Wikipedia is a frequent source for pre-training LLMs and the evaluated LLMs in this study have a knowledge cutoff date beyond April 2019 (as shown in Table 4), it can be inferred that the knowledge in these datasets

Answer Type	Abb	Template
Number	T1	How many gold medals did [country/region] win at the XXXIV Summer Olympic Games?
	T2	In the 25th FIFA World Cup, what was the final ranking of [country/region]?
	T3	How many children does [person] have?
	T4	How old was [person] in 2015?
Person	T5	Who won the bronze medal of [medal event] at the XXXIII Summer Olympic Games?
	T6	Who is the supreme leader of [country/region] in 2040?
	T7	In 2028, who served as the head coach of [country/region] national football team?
	T8	Who is [person]’s mom?
Time	T9	On which date was [person] born?
	T10	In what year did [person] die?
	T11	In what year did [person] graduate with the bachelor’s degree?
	T12	When was the wedding date for [person]?
Location	T13	Where was [person] born?
	T14	Where did [person] pass away?
	T15	Which university did [person] attend for the undergraduate studies?
	T16	Where was [person]’s wedding held?
Others	T17	What was the cause of [person]’s death?
	T18	What is the title of the debut album released by [person]?
	T19	What is the name of the first film directed by [person]
	T20	What is the occupation of [person]?

Table 1: Question templates used to create UnseenQA

is available to LLMs during their training. The creation of SeenQA involved a three-step process: 1) **Factoid Question Extraction**: exclude "why" questions, those with multiple answers, or answers exceeding five tokens. 2) **Time-Sensitive Question Removal**: use GPT-4-1106-PREVIEW (prompt in Table 5) to detect and remove questions with time-variant answers. 3) **Random Sampling**: randomly select 1,000 questions per source, discarding supporting context for a closed-book setting.

UnseenQA UnseenQA is a new QA dataset designed to ensure the LLMs tested in this study lack access to knowledge required to answer questions. It consists of 3,000 questions generated from 20 hand-written templates (150 questions per template), as detailed in Table 1, spanning five answer types: number, person, time, location, and others. Templates T1–T7 focus on future events with unknown answers at the time of writing, while the remaining templates involve fictional individuals whose names and details are not available online. The templates feature three placeholder types: 1) **Country/Region**: 150 names sourced from the National Olympic Committees listed on the [Wikipedia page](#). 2) **Medal Event**: 150 medal events from the [official programme of the Olympic games, Paris 2024](#). 3) **Person**: 150 randomly generated names from combinations of 100 first, middle, and last names, manually verified to lack online presence. The dataset was created on April 13, 2024. All LLMs studied were released before this date and

thus lacked access to the knowledge.

3.3 Evaluation on a Single Response

Uninformative We classify uninformative responses into three categories: **repetition**, **none**, and **unsure** (see Appendix C). **Repetition** refers to responses that repeatedly echo a specific string. We detect this using regular expressions and word frequency analysis. **None** includes responses that lack relevant content, such as empty strings or those merely repeating the question. **Unsure** applies to responses where the model says it is unable to answer or does not know. Examples include phrases like "I am not sure" or "I am just an AI", etc.

Correct A response is considered correct if it is an exact match with the ground truth or is similar enough as judged by GPT-4o. Following prior work (Sun et al., 2023), which reported a 98% agreement rate between ChatGPT and human evaluations, we adopt their evaluation prompt (detailed in Table 7 in Appendix D).

Consistency Score To compute $\text{Cons}(q, r)$, we set X_{MCQs} (total number of MCQs) to 20, and each MCQ includes question q and 5 options (the original response r , 3 random distractor options, and an ‘unsure’ option).

3.4 Prompts and Hyper-parameters

All LLMs were evaluated using greedy decoding (temperature 0 for GPT-3.5-TURBO) with a maximum of 100 new tokens. To provide a compre-

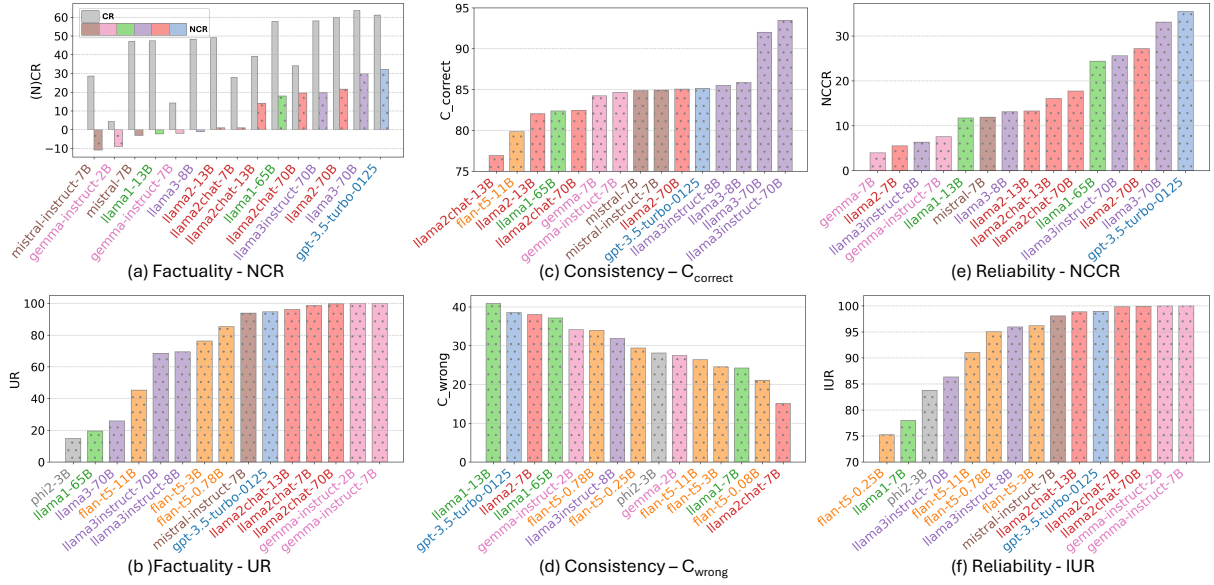


Figure 2: Top 15 LLMs ranked by (a) Net Correct Rate, (b) Uninformative Rate, (c) $C_{correct}$, (d) C_{wrong} , (d) Net Consistent Correct Rate, (e) Inconsistent/Uninformative Rate. Values are scaled by 100 (full results in Appendix E).

hensive evaluation, we experimented with three types of prompt settings: zero-shot, four-shot, and four-shot with two unsure shots. To avoid any bias introduced by fixed examples, we employed a dynamic few-shot method following the work of Nori et al. (2023). For further details of the prompt settings, refer to Appendix D.

4 Quantitative Results

We present results for all LLMs under different prompt settings in in Appendix E (Table 9: factuality, Table 10: consistency, and Table 11: reliability). LLM rankings based on different metrics are shown in Figure 6 and Figure 7, and also in Appendix E. In the remainder, we discuss our results in dynamic four-shot settings.

Which LLM is most factual? As shown in Figures 2 (a) and (b), GPT-3.5-TURBO performs best on seen knowledge, while GEMMA-INSTRUCT-7B excels at unseen knowledge. GPT-3.5-TURBO and LLAMA2CHAT-70B emerge as the most balanced models, demonstrating top performance across both seen and unseen knowledge. Notably, Figure 2 (a) highlights a limitation of CR, the standard metric for factuality. For instance, while LLAMA1-13B has a significantly higher CR than GEMMA-INSTRUCT-7B, their behavior differs in terms of wrong responses. LLAMA1-13B exhibits a 33.50% higher WR rate, leading to a lower NCR compared to GEMMA-INSTRUCT-7B.

Which LLM is most consistent? As shown in Figures 2 (c) and (d), LLAMA3INSTRUCT-70B is most consistent on questions it answers correctly, while LLAMA2CHAT-7B is least consistent on questions it answers incorrectly. Among these models, LLAMA3INSTRUCT-8B is most balanced, with a high $C_{correct}$ and low C_{wrong} .

Which LLM is most reliable? As shown in Figures 2 (e) and (f), GPT-3.5-TURBO excels at seen knowledge, while GEMMA-INSTRUCT-7B performs best on unseen knowledge. Among these models, GPT-3.5-TURBO demonstrates balanced performance, with a top NCCR and a high IUR. Note that a more factual LLM is not necessarily more reliable. For example, LLAMA3-INSTRUCT-8B ranks below the 15th in NCR and 13th in NCCR due to its relatively high $C_{correct}$ and low C_{wrong} .

5 Discussion

5.1 Correlation Analysis

Does an LLM with strong performance on seen knowledge tend to perform strong on unseen knowledge? We next examine whether there is a linear relationship between performance on seen and unseen knowledge. Performance on seen knowledge does not reliably predict performance on unseen knowledge in zero-shot and few-shot settings (without unsure shots). Table 2 (the first and second rows) shows the correlation values (Pearson’s ρ) between NCR and UR, and NCCR and

Comparisons	Zero-Shot	Four-Shot	Four-Two
NCR vs. UR	0.27	0.34	<u>0.62</u>
NCCR vs. IUR	-0.17	-0.12	<u>0.41</u>
$C_{correct}$ vs. C_{wrong}	<u>0.81</u>	<u>0.78</u>	<u>0.51</u>
NCR vs. $C_{correct}$	<u>0.60</u>	<u>0.41</u>	<u>0.37</u>
NCR vs. C_{wrong}	<u>0.48</u>	<u>0.64</u>	<u>0.41</u>
UR vs. $C_{correct}$	<u>0.43</u>	0.07	0.35
UR vs. C_{wrong}	<u>0.34</u>	0.05	<u>0.48</u>

Table 2: Pairwise correlation of LLM performance on different metrics under different prompt settings. The correlations are computed across all LLMs (26 data points). We report Pearson’s ρ , with underlined values indicating statistical significance ($p < 0.05$). Four-Two refers to the four-shot setting with two unsure shots.

IUR, revealing no statistically significant correlations in these settings. However, in the four-shot setting with two unsure shots, correlations across all metrics are significant. While seen knowledge performance does not transfer to unseen knowledge, specific prompt manipulations can improve these correlations (see the last column in Table 2).

Does an LLM with high consistency score on correctly answered questions tend to perform less consistent on wrongly answered questions?

As shown in Figure 11 and Table 10 in Appendix E, an LLM tends to be more consistent on questions it answers correctly than on those it answers wrongly. However, when comparing different LLMs, models with higher $C_{correct}$ also tend to have higher C_{wrong} , as indicated by the positive, significant correlation reported in Table 2 (the third row). This finding suggests that LLMs consistent with correct responses are also consistent with wrong responses, contradicting the expectation of high $C_{correct}$ and low C_{wrong} . This highlights a notable flaw in current models, which future work should address.

Does strong factuality performance correlate with strong consistency performance?

As shown in Table 2 (last four rows), there is a positive correlation between NCR/UR and $C_{correct}/C_{wrong}$, particularly in zero-shot settings. This suggests that LLMs with strong factuality performance tend to be confident in their responses, maintaining consistency whether those responses are correct or wrong.

5.2 Model Size, Fine-tuning and ICL Impact

How does model size affect LLMs performance?

As shown in Figure 3(a), *larger LLMs perform better on seen knowledge but worse on unseen knowledge*. As model size increases, both NCR

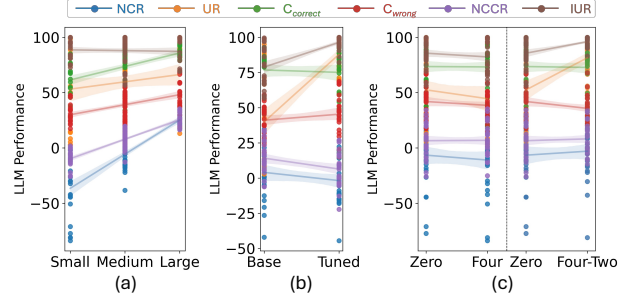


Figure 3: The impact of (a) model size, (b) fine-tuning, (c) ICL on LLM performance, measured with NCR, UR, $C_{correct}$, C_{wrong} , NCCR, and IUR. Different metrics are color-coded. See Appendix E for more detailed visualization. Values are scaled by 100.

(blue line) and NCCR (purple line) improve, indicating better performance on questions related to seen knowledge. However, while the UR (orange line) increases, the IUR (brown line) decreases. This suggests that larger models make fewer wrong responses on unseen knowledge but tend to give more consistent wrong responses. We also observe that *larger LLMs are more consistent, even with wrong responses*. Both $C_{correct}$ (green line) and C_{wrong} (red line) rise significantly with model size. While higher consistency in correct responses is desirable, the increase in consistency for wrong responses presents a risk. Larger models may confidently and consistently produce incorrect yet convincing information, heightening the potential for misinformation if not carefully managed.

How does fine-tuning affect LLMs performance?

Figure 3(b) suggests that *fine-tuning improves performance on unseen knowledge but degrades performance on seen knowledge*. After fine-tuning, both UR (orange line) and IUR (brown line) increase significantly, indicating improved handling of unseen knowledge. However, the decline in NCR (blue line) and NCCR (purple line) shows that fine-tuning reduces the model’s effectiveness with seen knowledge. We also see that *fine-tuning has no impact on consistency*. There is no notable change in C_{wrong} (red line) or $C_{correct}$ (green line) after fine-tuning which suggests that current instruction-tuning and RLHF methods do not improve LLM consistency.

How does ICL affect LLMs performance?

Figure 3(c) shows that *ICL does not improve performance on seen knowledge, but unsure shots enhance performance on unseen knowledge*. ICL does not increase NCR (blue line) or NCCR (purple line).

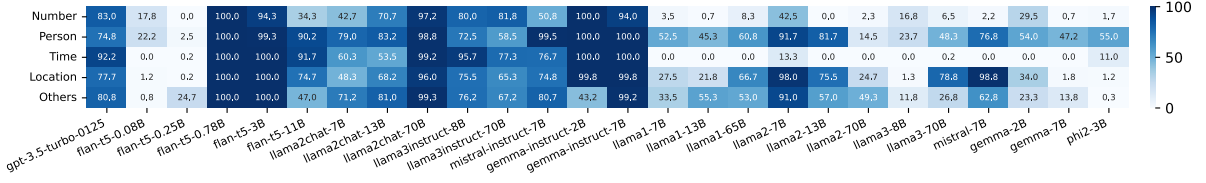


Figure 4: The impact of question type on LLM performance, measured by Uninformative Rate (UR) on unseen knowledge (scaled by 100). Questions are grouped by answer type. Higher values have darker shades.

ple line), which contrasts with previous findings on LLMs’ ICL ability (Brown et al., 2020; Chada and Natarajan, 2021; Touvron et al., 2023; Bai et al., 2023). Prior research largely focuses on the *correct rate* in few-shot settings without unsure shots. As shown in Table 9 and Figure 12 (Appendix E), while the four-shot setting significantly increases the correct rate, it also raises the wrong rate. Adding unsure shots helps reduce the wrong rate but also lowers the correct rate. For unseen knowledge, incorporating two unsure shots in the four-shot setting substantially improves UR (orange line) and IUR (brown line), indicating better handling of unknown questions. *ICL with unsure shots reduces consistency in wrong responses*. Adding two unsure shots in the four-shot setting decreases C_{wrong} (red line). Without unsure shots, there is no significant change in C_{wrong} (red line) or $C_{correct}$ (green line).

5.3 Behavior on Unseen Knowledge

How do LLMs perform on different types of questions about unseen knowledge? Figure 4 shows that base LLMs tend to overestimate their knowledge when answering numerical and temporal questions. Their UR scores are significantly lower for these types of queries, indicating a tendency to provide misleading responses even when they lack relevant knowledge.

6 Related Work

Petroni et al. (2019) first explored using pre-trained LMs as KBs, introducing LAMA and showing that BERT retains relational knowledge with precision as the metric. Roberts et al. (2020) evaluated knowledge storage and retrieval using natural language queries, measuring accuracy. Wang et al. (2021) fine-tuned BART with related passages to instill factual knowledge, assessing masked span recovery accuracy. He et al. (2024) trained T5 and LLaMA2 on Wikidata to evaluate large-scale knowledge retention via exact match and F1 scores. Sun et al.

(2023) tested LLMs on 18,000 fact-based QA pairs, reporting both accuracy and hallucination rates.

Many studies have highlighted the issue of factuality in current LLMs. To address this, new benchmarks have been proposed to assess factuality (Muhlgay et al., 2024; Zhao et al., 2024; Liu et al., 2024), although these still rely on correct rate as the primary metric. Several methods to enhance LLM factuality have also been introduced (Wang et al., 2022; Hase et al., 2024; Cohen et al., 2024; Qin et al., 2024). However, evaluating these improved models falls outside the scope of our experiments and is suggested for future work.

Research on consistency (Rajan et al., 2024; Sreekar et al., 2024; Saxena et al., 2024) has shown that LLMs often struggle with providing consistent responses. These works, however, do not differentiate between the consistency expectations for correctly and wrongly answered questions.

Compared to previous research, we propose a comprehensive framework to evaluate not only whether LLMs recall *seen* knowledge but also their ability to respond to *unseen* knowledge. In addition, we evaluate LLM consistency when answering questions about *identical* knowledge.

7 Conclusion

In this paper, we rethink the requirements for evaluating LLMs as KBs and propose criteria emphasizing factuality and consistency, and the combination thereof which we argue is an indicator of reliability. We proposed various metrics operationalizing these criteria and used them to assess LLM performance when answering questions pertaining to both seen and unseen knowledge. We evaluated 26 LLMs on our newly proposed SeenQA and UnseenQA datasets, and examined the impact of model size, fine-tuning, and ICL. Our experimental results highlight the critical need for continued research to develop more robust strategies that ensure both factuality and consistency, enabling LLMs to reliably function as KBs.

Limitations

First, due to budget constraints, we conducted in-depth evaluations on only a single closed-source LLM (GPT-3.5-TURBO). Nevertheless, this limitation does not diminish the contributions of our work. Our study includes a broad comparative analysis of 26 different LLMs, ensuring that the insights gained are comprehensive and not confined to any single model. Furthermore, the primary objective of our paper is to introduce a systematic framework for evaluating LLM-as-KB. This framework is universally applicable, offering value for the evaluation of a wide range of LLMs beyond those specifically analyzed in this study.

Second, our evaluation focuses primarily on factoid questions, which assess the models’ ability to recall specific factual knowledge rather than perform complex reasoning. Investigating how LLMs handle such complex queries remains an important direction for future research.

References

- Kim Allan Andersen and Daniele Pretolani. 2001. Easy cases of probabilistic satisfiability. *Annals of Mathematics and Artificial Intelligence*, 33:69–91.
- Jinze Bai, Shuai Bai, Yunfei Chu, Zeyu Cui, Kai Dang, Xiaodong Deng, Yang Fan, Wenbin Ge, Yu Han, Fei Huang, et al. 2023. Qwen technical report. *arXiv preprint arXiv:2309.16609*.
- Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. 2020. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901.
- Rakesh Chada and Pradeep Natarajan. 2021. [Few-shotQA: A simple framework for few-shot learning of question answering tasks using pre-trained text-to-text models](#). In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 6081–6090, Online and Punta Cana, Dominican Republic. Association for Computational Linguistics.
- Shiqi Chen, Yiran Zhao, Jinghan Zhang, I-Chun Chern, Siyang Gao, Pengfei Liu, and Junxian He. 2023. Felm: Benchmarking factuality evaluation of large language models. In *Thirty-seventh Conference on Neural Information Processing Systems Datasets and Benchmarks Track*.
- Roi Cohen, Eden Biran, Ori Yoran, Amir Globerson, and Mor Geva. 2024. Evaluating the ripple effects of knowledge editing in language models. *Transactions of the Association for Computational Linguistics*, 12:283–298.
- Yanai Elazar, Nora Kassner, Shauli Ravfogel, Abhilasha Ravichander, Eduard Hovy, Hinrich Schütze, and Yoav Goldberg. 2021. Measuring and improving consistency in pretrained language models. *Transactions of the Association for Computational Linguistics*, 9:1012–1031.
- Lovisa Hagström, Denitsa Saynova, Tobias Norlund, Moa Johansson, and Richard Johansson. 2023. The effect of scaling, retrieval augmentation and form on the factual consistency of language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 5457–5476.
- Peter Hase, Mohit Bansal, Been Kim, and Asma Ghan-deharioun. 2024. Does localization inform editing? surprising differences in causality-based localization vs. knowledge editing in language models. *Advances in Neural Information Processing Systems*, 36.
- Qiyuan He, Yizhong Wang, and Wenya Wang. 2024. Can language models act as knowledge bases at scale? *arXiv preprint arXiv:2402.14273*.
- Myeongjun Jang, Deuk Sin Kwon, and Thomas Lukasiewicz. 2022. Becel: Benchmark for consistency evaluation of language models. In *Proceedings of the 29th International Conference on Computational Linguistics*, pages 3680–3696.
- Mandar Joshi, Eunsol Choi, Daniel S Weld, and Luke Zettlemoyer. 2017. Triviaqa: A large scale distantly supervised challenge dataset for reading comprehension. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1601–1611.
- Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, et al. 2019. Natural questions: a benchmark for question answering research. *Transactions of the Association for Computational Linguistics*, 7:452–466.
- Stephanie Lin, Jacob Hilton, and Owain Evans. 2022. Truthfulqa: Measuring how models mimic human falsehoods. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3214–3252.
- Xiaozhe Liu, Feijie Wu, Tianyang Xu, Zhuo Chen, Yichi Zhang, Xiaoqian Wang, and Jing Gao. 2024. Evaluating the factuality of large language models using large-scale knowledge graphs. *arXiv preprint arXiv:2404.00942*.
- Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi. 2023. When not to trust language models: Investigating

- effectiveness of parametric and non-parametric memories. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9802–9822.
- Dor Muhlgay, Ori Ram, Inbal Magar, Yoav Levine, Nir Ratner, Yonatan Belinkov, Omri Abend, Kevin Leyton-Brown, Amnon Shashua, and Yoav Shoham. 2024. [Generating benchmarks for factuality evaluation of language models](#). In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 49–66, St. Julian’s, Malta. Association for Computational Linguistics.
- Harsha Nori, Yin Tat Lee, Sheng Zhang, Dean Carignan, Richard Edgar, Nicolo Fusi, Nicholas King, Jonathan Larson, Yuanzhi Li, Weishung Liu, et al. 2023. Can generalist foundation models outcompete special-purpose tuning? case study in medicine. *arXiv preprint arXiv:2311.16452*.
- Fabio Petroni, Tim Rocktäschel, Sebastian Riedel, Patrick Lewis, Anton Bakhtin, Yuxiang Wu, and Alexander Miller. 2019. Language models as knowledge bases? In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2463–2473.
- Jiaxin Qin, Zixuan Zhang, Chi Han, Manling Li, Pengfei Yu, and Heng Ji. 2024. Why does new knowledge create messy ripple effects in llms? *arXiv preprint arXiv:2407.12828*.
- Sai Sathiesh Rajan, Ezekiel Soremekun, and Sudipta Chattopadhyay. 2024. [Knowledge-based consistency testing of large language models](#). In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 10185–10196, Miami, Florida, USA. Association for Computational Linguistics.
- Adam Roberts, Colin Raffel, and Noam Shazeer. 2020. How much knowledge can you pack into the parameters of a language model? In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 5418–5426.
- Yash Saxena, Sarthak Chopra, and Arunendra Mani Tripathi. 2024. Evaluating consistency and reasoning capabilities of large language models. *arXiv preprint arXiv:2404.16478*.
- P Aditya Sreekar, Sahil Verma, Suransh Chopra, Abhishek Persad, Sarik Ghazarian, and Narayanan Sadagopan. 2024. [AXCEL: Automated eXplainable consistency evaluation using LLMs](#). In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 14943–14957, Miami, Florida, USA. Association for Computational Linguistics.
- Kai Sun, Yifan Ethan Xu, Hanwen Zha, Yue Liu, and Xin Luna Dong. 2023. Head-to-tail: How knowledgeable are large language models (llm)? aka will llms replace knowledge graphs? *arXiv preprint arXiv:2308.10168*.
- Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, et al. 2023. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*.
- Cunxiang Wang, Pai Liu, and Yue Zhang. 2021. Can generative pre-trained language models serve as knowledge bases for closed-book qa? In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 3241–3251.
- Haoyu Wang, Tianci Liu, Ruirui Li, Monica Xiao Cheng, Tuo Zhao, and Jing Gao. 2024a. [RoseLoRA: Row and column-wise sparse low-rank adaptation of pre-trained language model for knowledge editing and fine-tuning](#). In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 996–1008, Miami, Florida, USA. Association for Computational Linguistics.
- Mengru Wang, Ningyu Zhang, Ziwen Xu, Zekun Xi, Shumin Deng, Yunzhi Yao, Qishen Zhang, Linyi Yang, Jindong Wang, and Huajun Chen. 2024b. [Detoxifying large language models via knowledge editing](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3093–3118, Bangkok, Thailand. Association for Computational Linguistics.
- Weixuan Wang, Barry Haddow, and Alexandra Birch. 2024c. [Retrieval-augmented multilingual knowledge editing](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 335–354, Bangkok, Thailand. Association for Computational Linguistics.
- Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2022. Self-consistency improves chain of thought reasoning in language models. *arXiv preprint arXiv:2203.11171*.
- Yuxia Wang, Minghan Wang, Muhammad Arslan Manzoor, Georgi Georgiev, Rocktim Jyoti Das, and Preslav Nakov. 2024d. Factuality of large language models in the year 2024. *arXiv preprint arXiv:2402.02420*.
- Tian Yu, Shaolei Zhang, and Yang Feng. 2024. [Truth-aware context selection: Mitigating hallucinations of large language models being misled by untruthful contexts](#). In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 10862–10884, Bangkok, Thailand. Association for Computational Linguistics.
- Shaolei Zhang, Tian Yu, and Yang Feng. 2024a. [TruthX: Alleviating hallucinations by editing large language models in truthful space](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational*

Linguistics (Volume 1: Long Papers), pages 8908–8949, Bangkok, Thailand. Association for Computational Linguistics.

Xiaoying Zhang, Baolin Peng, Ye Tian, Jingyan Zhou, Lifeng Jin, Linfeng Song, Haitao Mi, and Helen Meng. 2024b. [Self-alignment for factuality: Mitigating hallucinations in LLMs via self-evaluation](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1946–1965, Bangkok, Thailand. Association for Computational Linguistics.

Yiran Zhao, Jinghan Zhang, I Chern, Siyang Gao, Pengfei Liu, Junxian He, et al. 2024. Felm: Benchmarking factuality evaluation of large language models. *Advances in Neural Information Processing Systems*, 36.

Danna Zheng, Danyang Liu, Mirella Lapata, and Jeff Z Pan. 2024. Trustscore: Reference-free evaluation of llm response trustworthiness. *arXiv preprint arXiv:2402.12545*.

A Datasets

SeenQA is composed of questions selected from the following open-sourced datasets:

1. Natural Questions (Kwiatkowski et al., 2019): This dataset includes questions sourced from web queries, each paired with a corresponding Wikipedia article containing the answer. The paper on Natural Questions was submitted to TACL in April 2018.
2. TriviaQA (Joshi et al., 2017): This dataset comprises questions from Quiz League websites, supplemented by web pages and Wikipedia searches that may contain the answer. The paper on TriviaQA was submitted to Arxiv in May 2017. For this project, we focus only on questions supported by Wikipedia.
3. PopQA (Mallen et al., 2023): This dataset targets long-tail entities. The authors used the Wikipedia dump from December 2018 in the retrieval augmented baseline, indicating that the knowledge in PopQA can be covered by the Wikipedia dump from that date.

Wikipedia is a common source in the pre-training data of large language models (LLMs). Comparing the knowledge cutoff dates provided in Table 4, we can deduce that the knowledge involved in these three datasets must have been seen during training by the LLMs used in our study.

B LLMs Used

Table 4 summarizes the LLMs used in our experiments.

C Uninformative Responses Examples

Table 3 provides some examples of uninformative responses. As illustrated in Figure 5 (fourth column), fine-tuned LLMs are able to explicitly acknowledge their lack of knowledge by answering ‘unsure’ to questions about unseen knowledge. In contrast, base LLMs often produce responses classified as ‘none’ or ‘repetition’ in the absence of unsure shots (contrast columns one and two with column three in Figure 5).

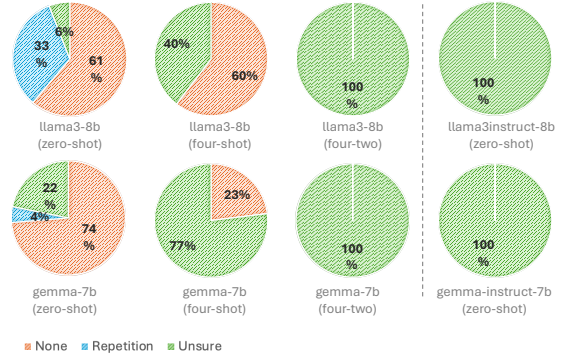


Figure 5: Distribution of uninformative responses given by LLMs to questions about unseen knowledge. We report results for the LLAMA3-8B, GEMMA-7B, and their fine-tuned models (fourth column) but observe similar trends on other models (omitted for the sake of brevity).

D Prompts Used

To provide a comprehensive evaluation, we experimented with three types of prompt settings: zero-shot, four-shot, and four-shot with two unsure shots. To avoid any bias introduced by fixed examples, we employed a dynamic few-shot method following the work of Nori et al. (2023). We collected two repositories, R_{seen} and R_{unseen} . R_{seen} includes 280 question-answer pairs about seen knowledge (200 from the unused data of PopQA and training data of Natural Questions and TriviaQA; 80 are generated using the templates in Table 1). R_{unseen} consists of 40 question-answer pairs about unseen knowledge, all generated using the templates in Table 1. We used `TEXT-EMBEDDING-3-SMALL` to embed the questions in the repositories and test questions as vector representations. For each test

Uninformative Type	Responses Examples
	1. (a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k) (l) (m) (n) (o) (p) (q) (r) (s) (t) (u) (v) (w) (x) (y) (z) (aa) (ab) (ac) (ad) (ae) (af) (
repetition	1. 2. 3. 4. 5. 6. 7. 8. 9. 10. 11. 12. 13. 14. 15. 16. 17. 18. 19. 20. 21. 22. 23. 24. 25. 26. 27. a swollen noah catherine cooper’s mom is _____.
none	‘ <image> unsure
unsure	i do not know. i’m just an ai, i don’t have access to real-time information or the ability to predict the future

Table 3: Uninformative Responses Examples

question under the four-shot setting, we retrieved its nearest four questions from R_{seen} . Under the four-shot with two unsure shots setting, we retrieved the nearest two questions from R_{seen} and two from R_{unseen} .

The prompt used to detect time-sensitive questions is shown in Table 5. The QA prompts under three different prompt settings are shown in Table 6. The prompt used to check whether an LLM’s response matches the ground-truth is shown in Table 7. The prompt used to generate distractors for the consistency test is shown in Table 8.

E Full Experimental Results

Table 9, Table 10, and Table 11 provides the detailed results of LLMs’ performance on factuality, consistency, and reliability respectively. Figure 6 shows the rankings of LLMs based on different metrics. Figure 7 compare different LLMs’ factuality, consistency, and reliability performance.

Figure 8, 9, and 10 provide detailed illustration of the impact of model size, fine-tuning, and ICL.

Figure 11 shows $C_{correct}$ and C_{wrongs} performance of LLMs ranked in $C_{correct}$.

Figure 12 shows correct rate and wrong ate under different prompt settings for LLMs ranked in correct rate under zero-shot.

Models	#Params	Type	Open Source	Fine-Tuning		Release Date	Pre-Training		
				IT	RLHF		Knowledge	# Token	Vocab
gpt-3.5-turbo-0125	Unknown	Dec-only	✗	✓	✓	25 Jan 2024	Sep 2021	-	-
Flan-T5	0.08B	Enc-Dec	✓	✓	✗	20 Oct 2022	<u>April 2019</u>	Unknown	32K
	0.25B	Enc-Dec	✓	✓	✗	20 Oct 2022	<u>April 2019</u>	Unknown	32K
	0.78B	Enc-Dec	✓	✓	✗	20 Oct 2022	<u>April 2019</u>	Unknown	32K
	3B	Enc-Dec	✓	✓	✗	20 Oct 2022	<u>April 2019</u>	Unknown	32K
	11B	Enc-Dec	✓	✓	✗	20 Oct 2022	<u>April 2019</u>	Unknown	32K
Llama1	7B	Dec-only	✓	✗	✗	27 Feb 2023	<u>Aug 2022</u>	1T	32K
	13B	Dec-only	✓	✗	✗	27 Feb 2023	<u>Aug 2022</u>	1T	32K
	65B	Dec-only	✓	✗	✗	27 Feb 2023	<u>Aug 2022</u>	1.4T	32K
Llama2	7B	Dec-only	✓	✗	✗	18 July 2023	Sep 2022	2T	32K
	13B	Dec-only	✓	✗	✗	18 July 2023	Sep 2022	2T	32K
	70B	Dec-only	✓	✗	✗	18 July 2023	Sep 2022	2T	32K
Llama2chat	7B	Dec-only	✓	✓	✓	18 July 2023	Sep 2022	2T	32K
	13B	Dec-only	✓	✓	✓	18 July 2023	Sep 2022	2T	32K
	70B	Dec-only	✓	✓	✓	18 July 2023	Sep 2022	2T	32K
Llama3	8B	Dec-only	✓	✗	✗	18 April 2024	Mar 2023	15T+	128K
	70B	Dec-only	✓	✗	✗	18 April 2024	Dec 2023	15T+	128K
Llama3Instruct	8B	Dec-only	✓	✓	✓	18 April 2024	Mar 2023	15T+	128K
	70B	Dec-only	✓	✓	✓	18 April 2024	Dec 2023	15T+	128K
Mistral	7B	Dec-only	✓	✗	✗	27 Sep 2023	Unknown	Unknown	32K
Mistral-Instruct	7B	Dec-only	✓	✓	✗	27 Sep 2023	Unknown	Unknown	32K
Gemma	2B	Dec-only	✓	✗	✗	21 Feb 2024	Unknown	3T	256K
	7B	Dec-only	✓	✗	✗	21 Feb 2024	Unknown	6T	256K
Gemma-Instruct	2B	Dec-only	✓	✓	✓	21 Feb 2024	Unknown	3T	256K
	7B	Dec-only	✓	✓	✓	21 Feb 2024	Unknown	6T	256K
Phi2	3B	Dec-only	✓	✗	✗	12 Dec 2023	Unknown	1.4T	50K

Table 4: Summary of LLMs used in our experiments. ‘IT’ denotes Instruction Tuning, and ‘RLHF’ refers to Reinforcement Learning from Human Feedback. ‘Knowledge’ indicates the knowledge cutoff date. Underlined dates were not explicitly provided by the authors but extrapolated from the datasets used for LLM training. Flan-T5’s base model is T5 version 1.1 pre-trained on the C4 dataset, filtered from web-extracted text in April 2019. Llama 1’s pre-training data includes Wikipedia dumps from June to August 2022.

Prompt for detecting time-sensitive questions

INSTRUCTION: Please provide the index of questions whose answers change yearly. Just return the index without explanations.

Here is the list of questions:

1. Who is the most paid player in EPL?
2. What is the capital of Louisiana?
3. Who won the Nobel Peace Prize in 2009?
4. What is the latest model of the iPhone currently available?

Index:

1, 4

Here is the list of questions:

[question placeholder]

Index:

Table 5: The prompt for detecting time-sensitive questions

QA prompt in zero-shot
INSTRUCTION: Please answer knowledge-related questions directly. Note: Please do not give anything other than the answer; Say "unsure" if you do not know.
QUESTION: [question placeholder] ANSWER:
QA prompt in four-shot
INSTRUCTION: Please answer knowledge-related questions directly. Note: Please do not give anything other than the answer; Say "unsure" if you do not know.
QUESTION: [question example 1 from R_{seen}] ANSWER: [answer 1]
QUESTION: [question example 2 from R_{seen}] ANSWER: [answer 2]
QUESTION: [question example 3 from R_{seen}] ANSWER: [answer 3]
QUESTION: [question example 4 from R_{seen}] ANSWER: [answer 4]
QUESTION: [question placeholder] ANSWER:
QA prompt in four-shot with few unsure shot
INSTRUCTION: Please answer knowledge-related questions directly. Note: Please do not give anything other than the answer; Say "unsure" if you do not know.
QUESTION: [question example 1 from R_{seen}] ANSWER: [answer 1]
QUESTION: [question example 2 from R_{seen}] ANSWER: [answer 2]
QUESTION: [question example 3 from R_{unseen}] ANSWER: unsure
QUESTION: [question example 4 from R_{unseen}] ANSWER: unsure
QUESTION: [question placeholder] ANSWER:

Table 6: The question answering prompt format. The shots are selected from repositories, R_{seen} and R_{unseen} . The order of shots is random. For the MCQ tests in consistency experiments, we edit the instruction line to *INSTRUCTION: Please answer knowledge-related multi-choice questions directly. Note: Please do not give anything other than the appropriate option (A, B, C, D or E); choose the option indicating "unsure" if you do not know.*

Prompt for check whether an answer matches the ground truth for the question

INSTRUCTION: You need to check whether the prediction of a question-answering system to a question is correct. You should make the judgment based on a list of ground truth answers provided to you. Your response should be "yes" if the prediction is correct or "no" if the prediction is wrong.

Question: Who authored The Taming of the Shrew (published in 2002)?

Ground truth: ["William Shakespeare", "Roma Gill"]

prediction: W Shakespeare

Correctness: yes

Question: What country is Maharashtra Metro Rail Corporation Limited located in?

Ground truth: ["India"]

prediction: Maharashtra

Correctness: no

Question: Edward Tise (known for Full Metal Jacket (1987)) is in what department?

Ground truth: ["sound department"]

Prediction: 2nd Infantry Division, United States Army

Correctness: no

Question: Which era did Michael Oakeshott belong to?

Ground truth: ["20th-century philosophy"]

prediction: 20th century.

Correctness: yes

Question: [quesetion placeholder]

Ground truth: [ground truth placeholder]

prediction: [LLM's answer placeholder]

Correctness:

Table 7: The prompt used to check whether an LLM's answer matches the ground truth for the question

Prompt for generating distractors

INSTRUCTION: For the given question-answer pair, provide 20 different distractors that are similar yet distinct from the given answer. Note: Seperate the 20 distractors with a special token "[SEP]".

Q: Who was the President of the United States in 2010?

A: Barack Obama

Distractors: George W. Bush [SEP] Bill Clinton [SEP] Ronald Reagan [SEP] Donald Trump [SEP] Jimmy Carter [SEP] George H.W. Bush [SEP] Richard Nixon [SEP] Gerald Ford [SEP] Lyndon B. Johnson [SEP] John F. Kennedy [SEP] Dwight D. Eisenhower [SEP] Harry S. Truman [SEP] Franklin D. Roosevelt [SEP] Herbert Hoover [SEP] Calvin Coolidge [SEP] Woodrow Wilson [SEP] William Howard Taft [SEP] Theodore Roosevelt [SEP] William McKinley [SEP] Grover Cleveland

Q: What is the name of the first cloned sheep?

A: the first cloned sheep is dolly.

Distractors: the first cloned sheep is Polly [SEP] the first cloned sheep is Molly [SEP] the first cloned sheep is Holly [SEP] the first cloned sheep is Bella [SEP] the first cloned sheep is Daisy [SEP] the first cloned sheep is Lily [SEP] the first cloned sheep is Rosie [SEP] the first cloned sheep is Millie [SEP] the first cloned sheep is Ellie [SEP] the first cloned sheep is Sally [SEP] the first cloned sheep is Tilly [SEP] the first cloned sheep is Nelly [SEP] the first cloned sheep is Jolly [SEP] the first cloned sheep is Betty [SEP] the first cloned sheep is Annie [SEP] the first cloned sheep is Lucy [SEP] the first cloned sheep is Maggie [SEP] the first cloned sheep is Cindy [SEP] the first cloned sheep is Penny [SEP] the first cloned sheep is Ginny

Q: [QUESTION]

A: [ANSWER]

Distractors:

Table 8: The prompt used to generate distractors for consistency tests.

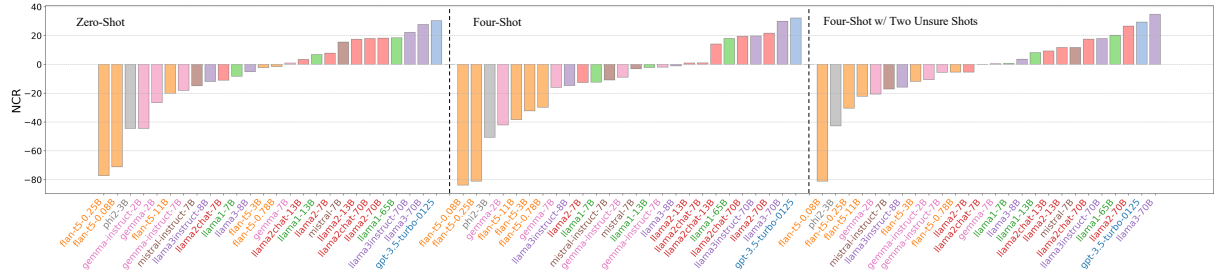
Model	Params	zero-shot				four-shot				four-shot-2			
		Seen		Unseen		Seen		Unseen		Seen		Unseen	
		WR (↓)	CR (↑)	NCR (↑)	UR (↑)	WR (↓)	CR (↑)	NCR (↑)	UR (↑)	WR (↓)	CR (↑)	NCR (↑)	UR (↑)
GPT-3.5 Turbo	Unknown	30.40	60.73	30.33	81.70	28.97	61.17	32.20	94.70	27.93	57.30	29.37	99.37
Flan-T5	0.08B	73.03	1.83	-71.20	8.40	85.53	1.63	-83.90	2.77	82.57	1.43	-81.13	34.63
	0.25B	82.77	5.47	-77.30	5.50	86.43	5.27	-81.17	2.70	32.67	2.23	-30.43	74.67
	0.78B	3.70	2.13	-1.57	100.00	37.80	8.00	-29.80	85.30	9.60	4.17	-5.43	99.90
	3B	9.70	7.50	-2.20	98.73	46.00	13.67	-32.33	76.27	23.03	11.23	-11.80	99.73
	11B	40.97	20.77	-20.20	67.57	60.63	22.23	-38.40	45.27	42.37	20.20	-22.17	90.00
Llama 1	7B	43.93	35.67	-8.27	23.40	54.47	42.10	-12.37	4.73	27.57	28.27	0.70	54.50
	13B	34.33	41.13	6.80	24.63	49.67	47.43	-2.23	4.10	27.37	35.53	8.17	69.37
	65B	28.40	46.87	18.47	37.77	39.77	57.73	17.97	19.47	14.63	34.87	20.23	90.10
Llama 2	7B	23.73	31.50	7.77	67.30	54.63	42.03	-12.60	5.33	35.53	30.10	-5.43	66.60
	13B	23.67	41.07	17.40	42.83	48.20	49.17	0.97	6.03	27.40	39.07	11.67	71.23
	70B	37.17	55.33	18.17	18.17	38.20	59.83	21.63	13.03	19.50	46.03	26.53	95.10
Llama2chat	7B	47.37	36.33	-11.03	60.30	26.87	27.90	1.03	98.60	23.43	23.33	-0.10	99.73
	13B	37.87	41.27	3.40	71.30	25.03	39.13	14.10	96.13	32.00	41.27	9.27	94.90
	70B	29.53	47.50	17.97	98.10	14.57	34.10	19.53	99.63	15.13	32.60	17.47	100.00
Llama3	8B	50.20	45.13	-5.07	10.73	49.27	48.23	-1.03	6.50	32.77	36.33	3.57	89.03
	70B	27.87	55.53	27.67	32.13	33.70	63.60	29.90	25.93	18.87	53.60	34.73	87.63
Llama3Instruct	8B	53.93	42.03	-11.90	79.97	54.00	39.27	-14.73	69.43	54.60	38.73	-15.87	78.73
	70B	36.80	59.03	22.23	70.03	38.40	58.10	19.70	68.47	38.90	56.80	17.90	88.60
Mistral	7B	24.00	39.47	15.47	48.13	50.13	47.07	-3.07	13.57	24.70	36.37	11.67	81.73
Mistral-Instruct	7B	44.77	29.90	-14.87	76.50	39.53	28.63	-10.90	93.80	46.63	29.47	-17.17	79.13
Gemma	2B	51.20	24.63	-26.57	28.17	69.17	27.07	-42.10	2.77	39.37	18.67	-20.70	56.07
	7B	38.80	39.73	0.93	12.70	56.50	40.53	-15.97	8.67	30.77	31.23	0.47	68.93
Gemma-Instruct	2B	53.80	9.27	-44.53	88.60	13.27	4.30	-8.97	99.93	14.40	3.77	-10.63	99.30
	7B	37.13	19.03	-18.10	98.60	16.17	14.20	-1.97	99.97	19.13	13.60	-5.53	99.93
Phi2	3B	65.97	21.43	-44.53	13.83	72.10	21.40	-50.70	14.77	62.07	19.33	-42.73	50.20

Table 9: Factuality performance (Values are scaled by 100).

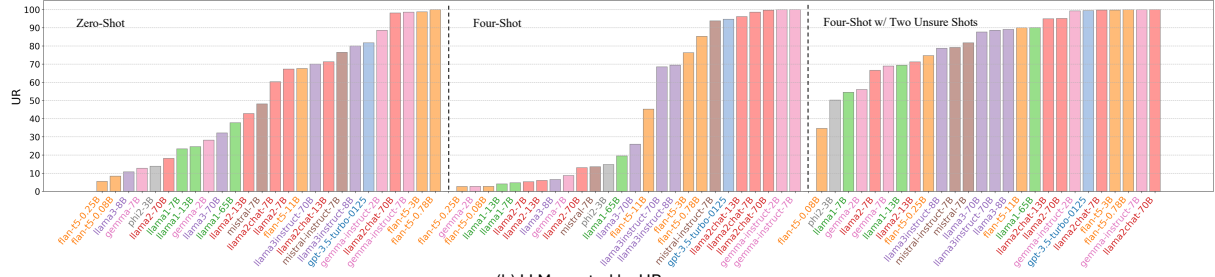
Model	Params	zero-shot				four-shot				four-shot-2			
		<i>C_{strong}</i> (↓)	<i>C_{strong}</i> (↓)	<i>C_{strong}</i> (↓)	<i>C_{correct}</i> (↑)	<i>C_{strong}</i> (↓)	<i>C_{strong}</i> (↓)	<i>C_{strong}</i> (↓)	<i>C_{correct}</i> (↑)	<i>C_{strong}</i> (↓)	<i>C_{strong}</i> (↓)	<i>C_{strong}</i> (↓)	<i>C_{correct}</i> (↑)
GPT-3.5 Turbo	-	61.79	23.65	42.72	87.10	57.43	19.62	38.53	85.16	48.56	33.68	41.12	79.68
Flan-T5	0.08B	14.49	20.56	17.53	28.64	16.53	25.62	21.07	47.45	18.44	25.87	22.15	47.21
	0.25B	35.33	26.31	30.82	62.29	33.36	25.44	29.40	69.40	35.34	22.80	29.07	75.90
	0.78B	45.68	-	45.68	85.23	34.16	33.72	33.94	75.12	42.99	35.00	38.99	82.64
	3B	45.03	25.26	35.15	84.20	33.07	16.05	24.56	76.85	37.13	35.62	36.38	80.96
	11B	41.62	15.38	28.50	80.43	36.40	16.40	26.40	79.84	40.74	15.07	27.90	80.61
Llama 1	7B	25.01	21.70	23.36	37.43	25.37	23.10	24.23	39.65	23.07	20.89	21.98	34.17
	13B	35.13	16.41	25.77	59.11	45.60	36.20	40.90	72.49	48.54	25.45	36.99	73.74
	65B	58.06	33.84	45.95	83.38	58.35	37.12	47.73	82.38	63.63	16.63	40.13	83.64
Llama 2	7B	26.68	9.37	18.03	50.02	41.51	34.56	38.03	67.07	41.94	25.26	33.60	67.29
	13B	62.02	35.69	48.86	83.08	56.12	45.05	50.58	82.05	58.55	31.89	45.22	83.65
	70B	63.13	37.52	50.33	84.36	62.07	35.37	48.72	85.06	52.84	6.43	29.63	79.10
Llama2chat	7B	43.66	15.63	29.64	61.23	17.69	12.50	15.09	20.15	19.82	20.62	20.22	17.99
	13B	55.79	32.88	44.33	74.62	56.11	28.97	42.54	76.92	52.23	27.39	39.81	77.52
	70B	73.61	59.65	66.63	88.71	71.28	30.91	51.10	82.45	67.82	-	67.82	81.88
Llama3	8B	64.17	48.52	56.35	86.50	57.43	35.72	46.58	85.85	37.86	9.51	23.69	78.80
	70B	76.82	41.82	59.32	92.86	75.47	41.62	58.55	92.00	64.67	9.43	37.05	86.07
Llama3Instruct	8B	53.08	29.74	41.41	88.86	50.43	13.25	31.84	85.51	37.34	3.80	20.57	79.64
	70B	78.14	59.26	68.70	94.24	74.80	43.32	59.06	93.47	67.25	28.17	47.71	93.03
Mistral	7B	56.79	26.70	41.75	84.15	55.94	37.50	46.72	84.87	51.19	13.06	32.13	83.21
Mistral-Instruct	7B	65.84	31.48	48.66	86.09	62.93	30.99	46.96	84.92	61.64	27.63	44.63	84.21
Gemma	2B	30.18	26.26	28.22	37.77	26.93	27.99	27.46	45.90	28.41	22.11	25.26	47.42
	7B	62.41	47.24	54.83	86.17	53.39	41.94	47.66	84.22	52.49	22.35	37.42	85.89
Gemma-Instruct	2B	51.65	42.72	47.19	59.64	53.23	15.00	34.11	54.11	50.51	44.52	47.52	49.51
	7B	81.92	51.79	66.85	92.43	72.34	30.00	51.17	84.64	80.70	30.00	55.35	90.56
Phi2	3B	30.09	18.27	24.18	54.92	37.21	19.02	28.11	67.34	38.48	15.78	27.13	68.16

Table 10: Consistency performance (Values are scaled by 100).

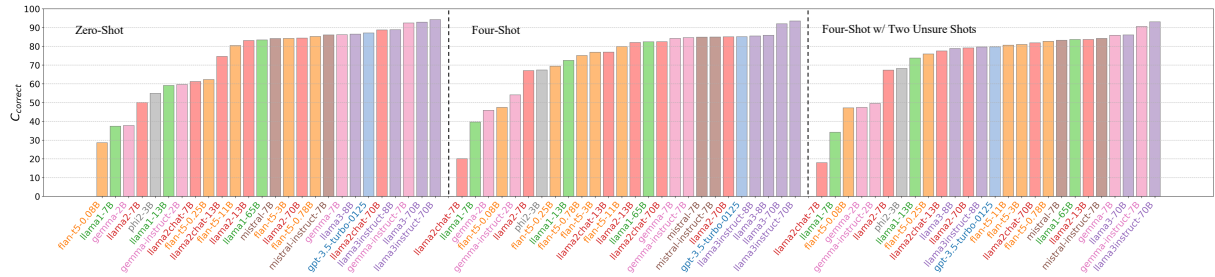
Model	Params	zero-shot				four-shot				four-shot-2			
		Seen		Unseen		Seen		Unseen		Seen		Unseen	
		CWR (↓)	CCR (↑)	NCCR (↑)	IUR (↑)	CWR (↓)	CCR (↑)	NCCR (↑)	IUR (↑)	CWR (↓)	CCR (↑)	NCCR (↑)	IUR (↑)
GPT-3.5 Turbo	-	18.78	52.90	34.11	95.67	16.64	52.09	35.45	98.96	13.56	45.66	32.09	99.79
Flan-T5	0.08B	10.58	0.52	-10.06	81.17	14.13	0.77	-13.36	75.09	15.22	0.68	-14.55	83.09
	0.25B	29.25	3.41	-25.84	75.13	28.83	3.66	-25.17	75.25	11.54	1.69	-9.85	94.22
	0.78B	1.69	1.82	0.13	100.00	12.91	6.01	-6.90	95.04	4.13	3.45	-0.68	99.97
	3B	4.37	6.32	1.95	99.68	15.21	10.51	-4.70	96.19	8.55	9.09	0.54	99.90
	11B	17.05	16.70	-0.35	95.01	22.07	17.75	-4.32	91.02	17.26	16.28	-0.98	98.49
Llama 1	7B	10.99	13.35	2.36	83.38	13.82	16.69	2.88	77.99	6.36	9.66	3.30	90.50
	13B	12.06	24.31	12.25	87.63	22.65	34.38	11.73	65.29	13.28	26.20	12.91	92.20
	65B	16.49	39.08	22.59	78.94	23.21	47.56	24.35	70.10	9.31	29.16	19.85	98.35
Llama 2	7B	6.37	15.76	9.39	96.94	22.68	28.19	5.51	67.28	14.90	20.25	5.35	91.56
	13B	14.68	34.12	19.44	79.60	27.05	40.34	13.29	57.67	16.04	32.68	16.64	90.83
	70B	23.47	46.68	23.21	69.30	23.71	50.89	27.18	69.24	10.30	36.41	26.11	99.69
Llama2chat	7B	20.68	22.24	1.56	93.80	4.75	5.62	0.87	99.83	4.64	4.20	-0.45	99.94
	13B	21.13	30.80	9.67	90.56	14.04	30.10	16.05	98.88	16.71	31.99	15.28	98.60
	70B	21.74	42.14	20.40	98.87	10.39	28.12	17.73	99.89	10.26	26.69	16.43	100.00
Llama3	8B	32.22	39.04	6.82	56.69	28.30	41.41	13.11	66.60	12.41	28.63	16.22	98.96
	70B	21.41	51.57	30.15	71.62	25.43	58.52	33.08	69.17	12.20	46.13	33.93	98.83
Llama3Instruct	8B	28.62	37.35	8.72	94.04	27.23	33.58	6.34	95.95	20.39	30.85	10.46	99.19
	70B	28.76	55.63	26.88	82.24	28.72	54.31	25.59	86.34	26.16	52.84	26.68	96.79
Mistral	7B	13.63	33.21	19.58	86.15	28.04	39.95	11.90	67.59	12.65	30.26	17.62	97.61
Mistral-Instruct	7B	29.48	25.74	-3.74	92.60	24.88	24.31	-0.56	98.08	28.74	24.82	-3.92	94.24
Gemma	2B	15.45	9.30	-6.15	81.14	18.62	12.42	-6.20	72.78	11.19	8.85	-2.33	90.29
	7B	24.22	34.24	10.02	58.76	30.16	34.14	3.97	61.70	16.15	26.82	10.67	93.06
Gemma-Instruct	2B	27.79	5.53	-22.26	95.13	7.06	2.33	-4.74	99.99	7.27	1.87	-5.41	99.98
	7B	30.42	17.59	-12.83	99.27	4.46	12.02	7.55	99.99	15.44	12.32	-3.12	99.69
Phi2	3B	19.85	11.77	-8.08	84.25	26.83	14.41	-12.42	83.79	23.88	13.17	-10.71	92.14



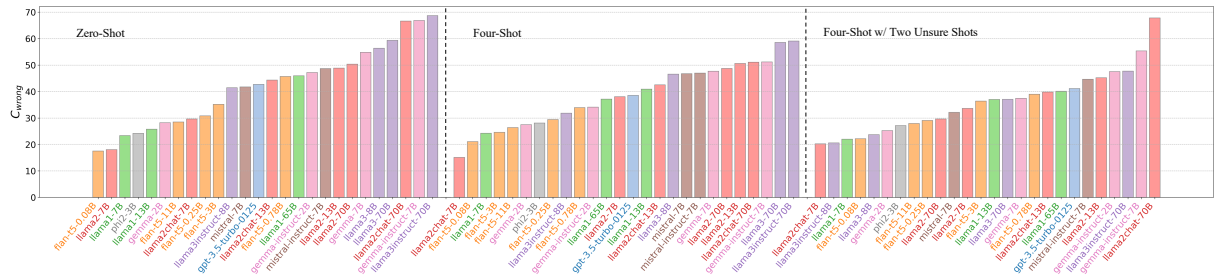
(a) LLMs sorted by NCR



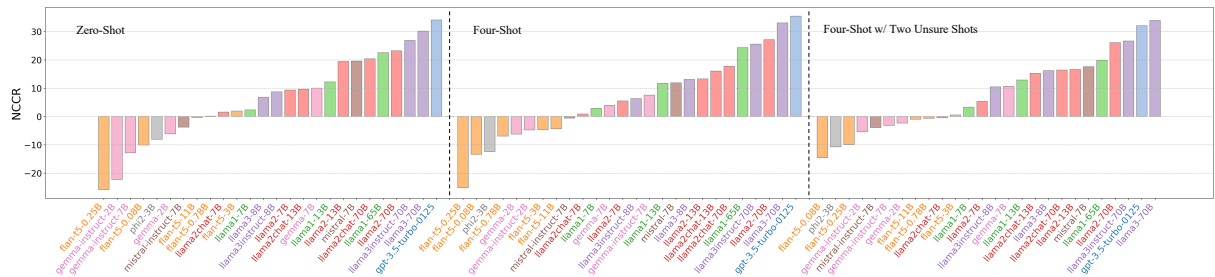
(b) LLMs sorted by UR



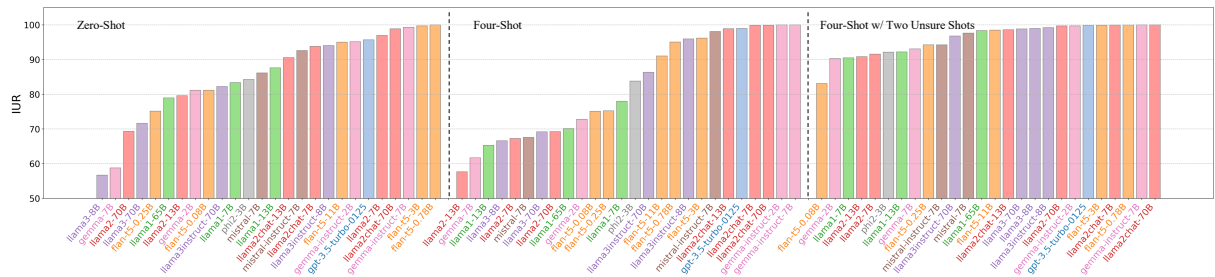
(c) LLMs sorted by $C_{correct}$



(d) LLMs sorted by C_{wrong}

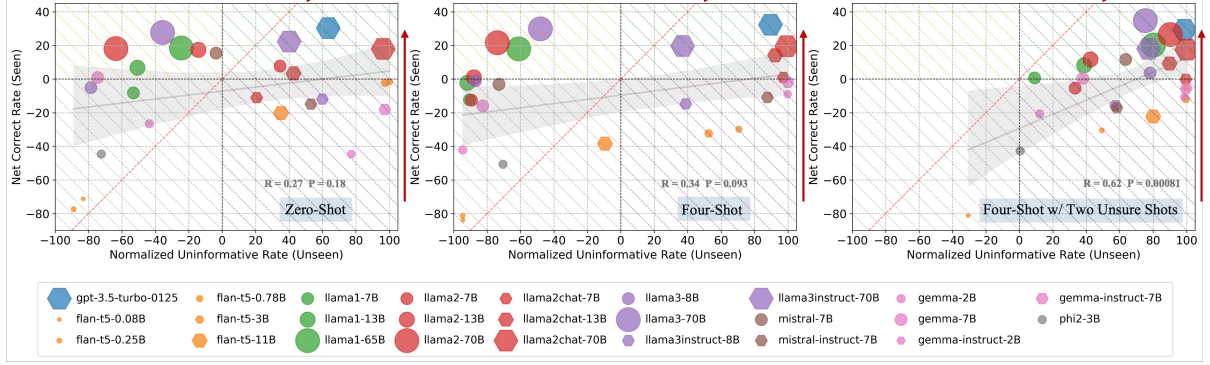


(e) LLMs sorted by NCCR

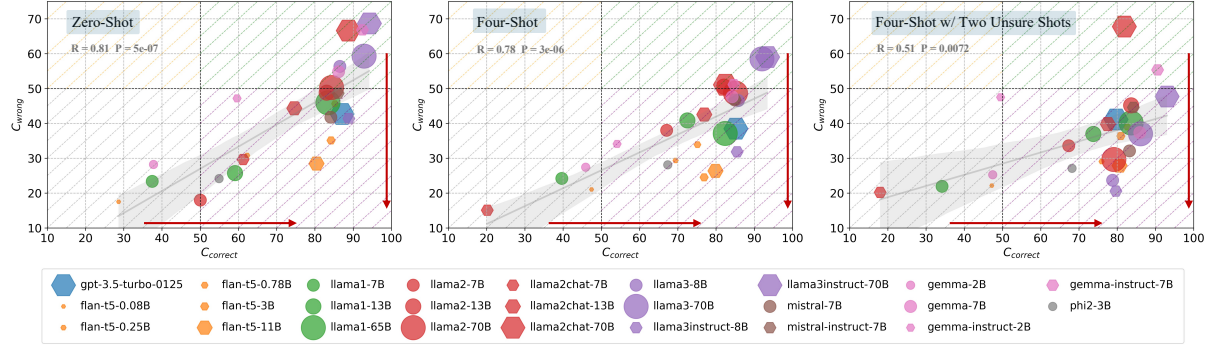


(f) LLMs sorted by IUR

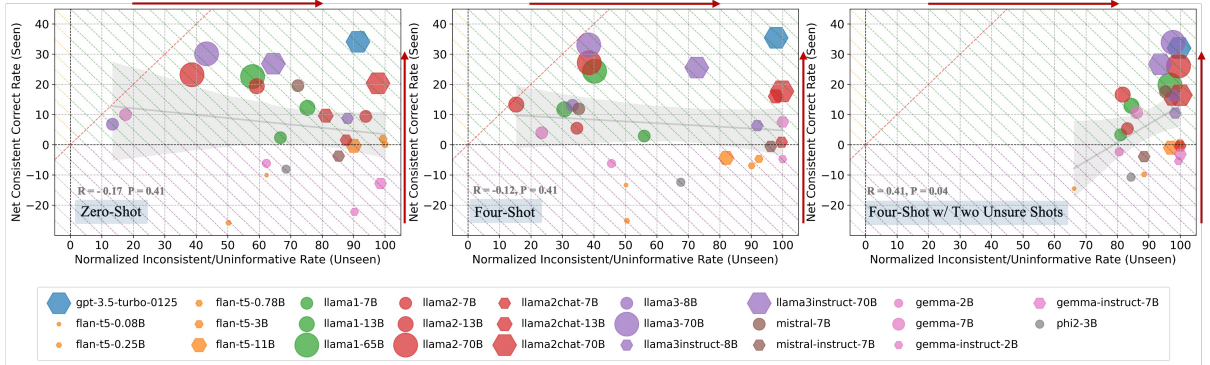
Figure 6: Ranking of LLMs based on different Metrics.



(a) Factuality performance on seen knowledge vs. unseen knowledge. R is the Pearson correlation coefficient. When $P < 0.05$, R is statistically significant. The red line is $y=x$. The LLMs above the red line perform better on seen knowledge. The LLMs below the red line perform better on unseen knowledge. LLMs closer to the top right corner are more factual (higher NCR and higher UR).



(b) Consistency performance on on wrong responses vs. correct responses. R is the Pearson correlation coefficient. When $P < 0.05$, R is statistically significant. LLMs closer to the bottom right corner are better in consistency (higher $C_{correct}$ and lower $C_{correct}$).



(c) Reliability performance on seen knowledge vs. unseen knowledge. R is the Pearson correlation coefficient. When $P < 0.05$, R is statistically significant. The red line is $y=x$. The LLMs above the red line perform better on seen knowledge. The LLMs below the red line perform better on unseen knowledge. LLMs closer to the top right corner are more reliable (higher NCR and higher UR).

Figure 7: Visualization of LLMs' factuality, consistency and reliability performance.

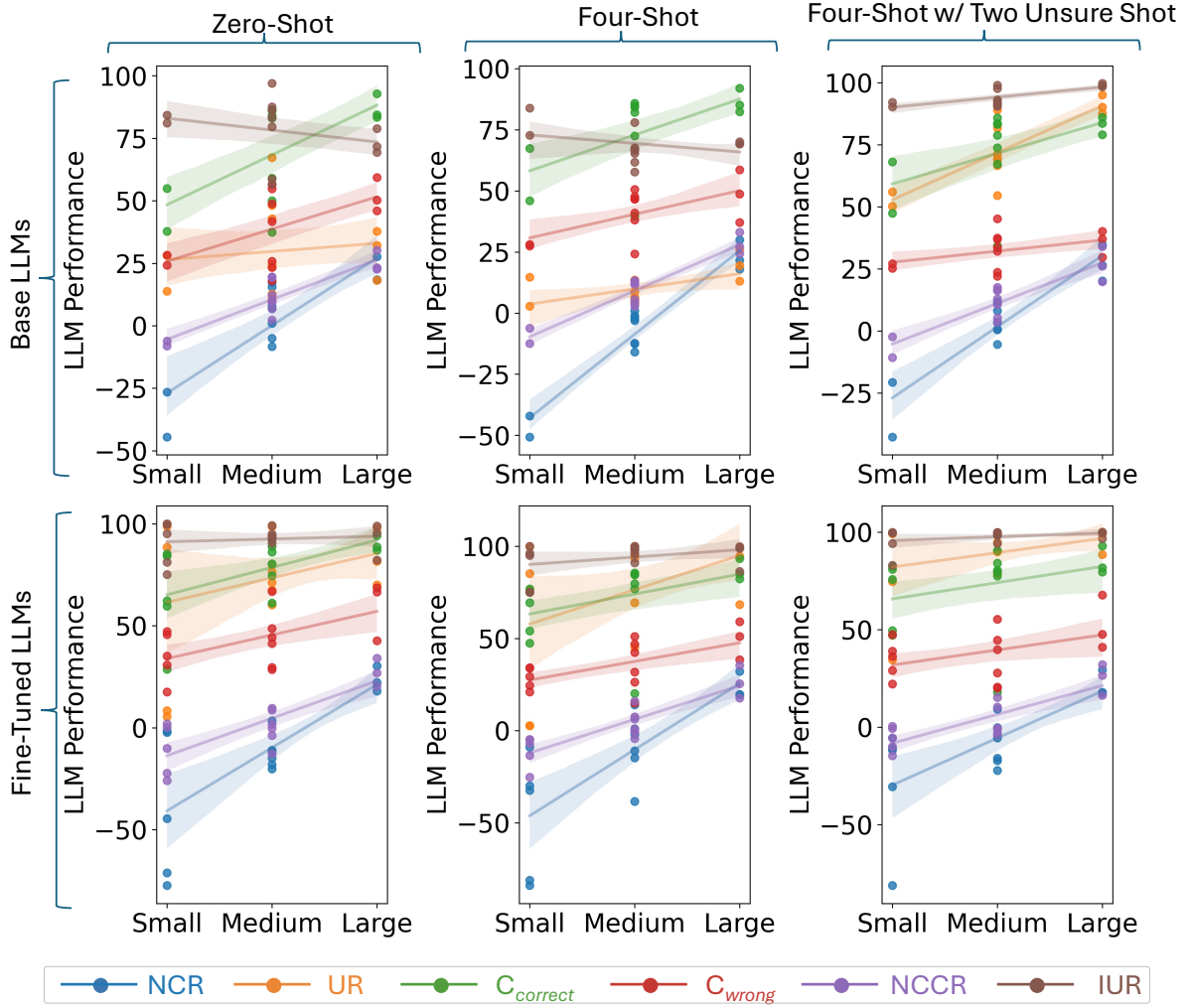


Figure 8: The impact of model size on LLM performance, measured with NCR, UR, $C_{correct}$, C_{wrong} , NCCR, and IUR (values are scaled by 100). Different metrics are color-coded. LLMs are shown in three sizes, small, medium, and large and are grouped into ‘base’ and fine-tuned ones.

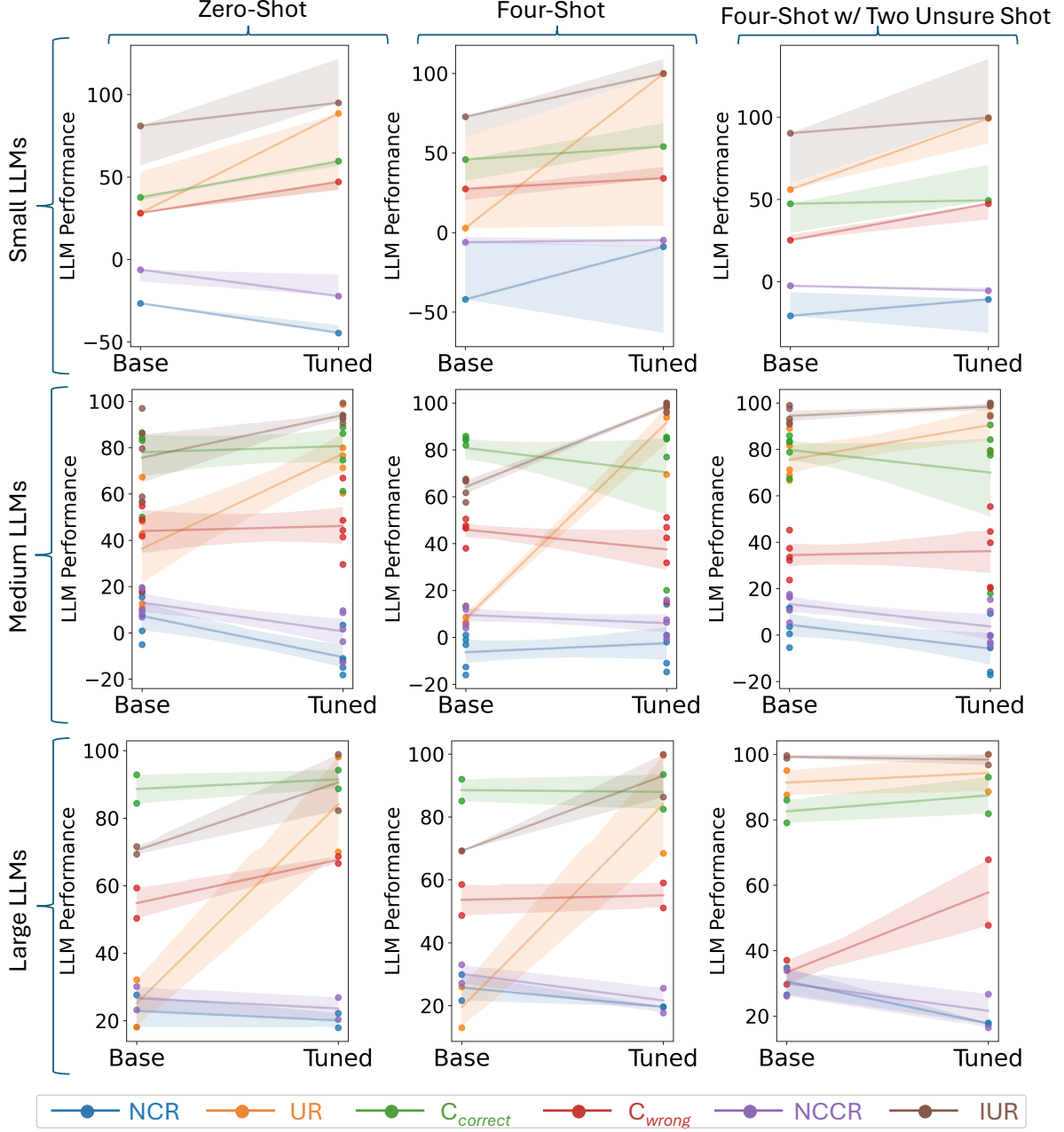


Figure 9: The impact of fine-tuning on LLM performance, measured with NCR, UR, $C_{correct}$, C_{wrong} , NCCR, and IUR (values are scaled by 100). Different metrics color-coded. This analysis only considers the performance of Llama2, Llama3, Mistral, and Gemma as these families include both base LLMs and fine-tuned versions. Models are shown in three sizes, small, medium, and large.

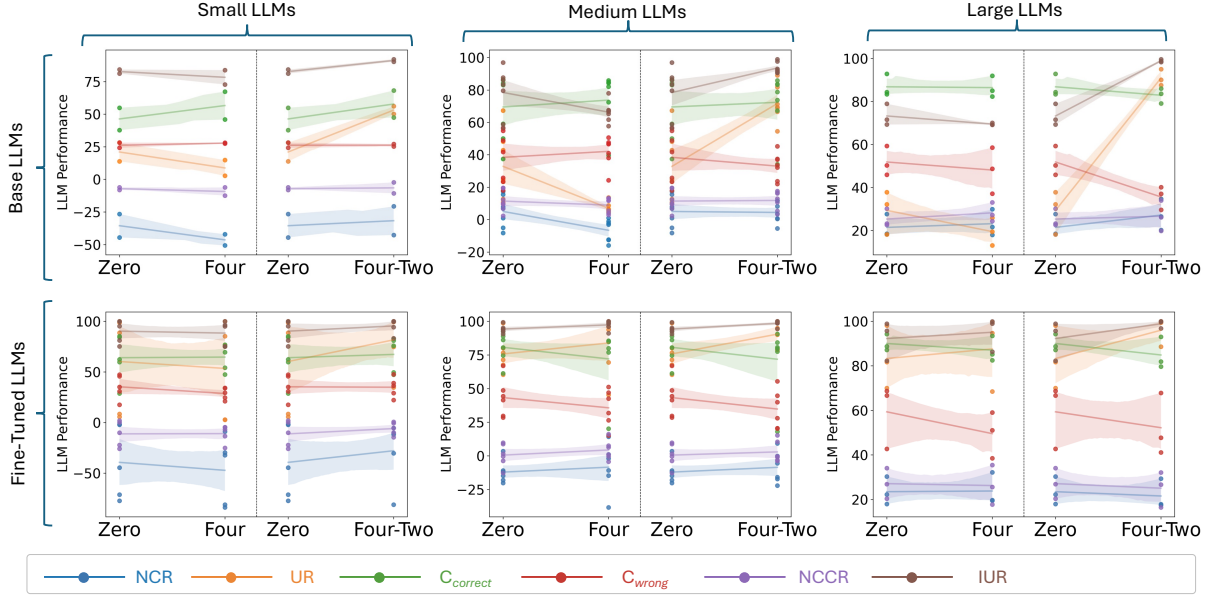


Figure 10: The impact of ICL on LLM performance, measured with NCR, UR, $C_{correct}$, C_{wrong} , NCCR, and IUR (values are scaled by 100). Different metrics are color-coded. We compare zero-shot and four-shot settings; and zero-shot against four-shot with two unsure shots. LLMs (‘base’ and fine-tuned ones) are in three sizes, small, medium, and large.

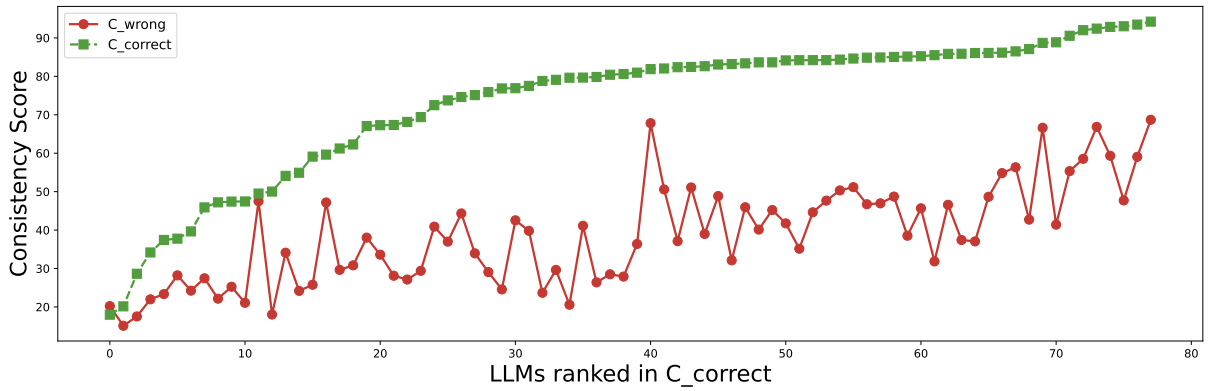


Figure 11: $C_{correct}$ and C_{wrong} performance of LLMs ranked in $C_{correct}$.

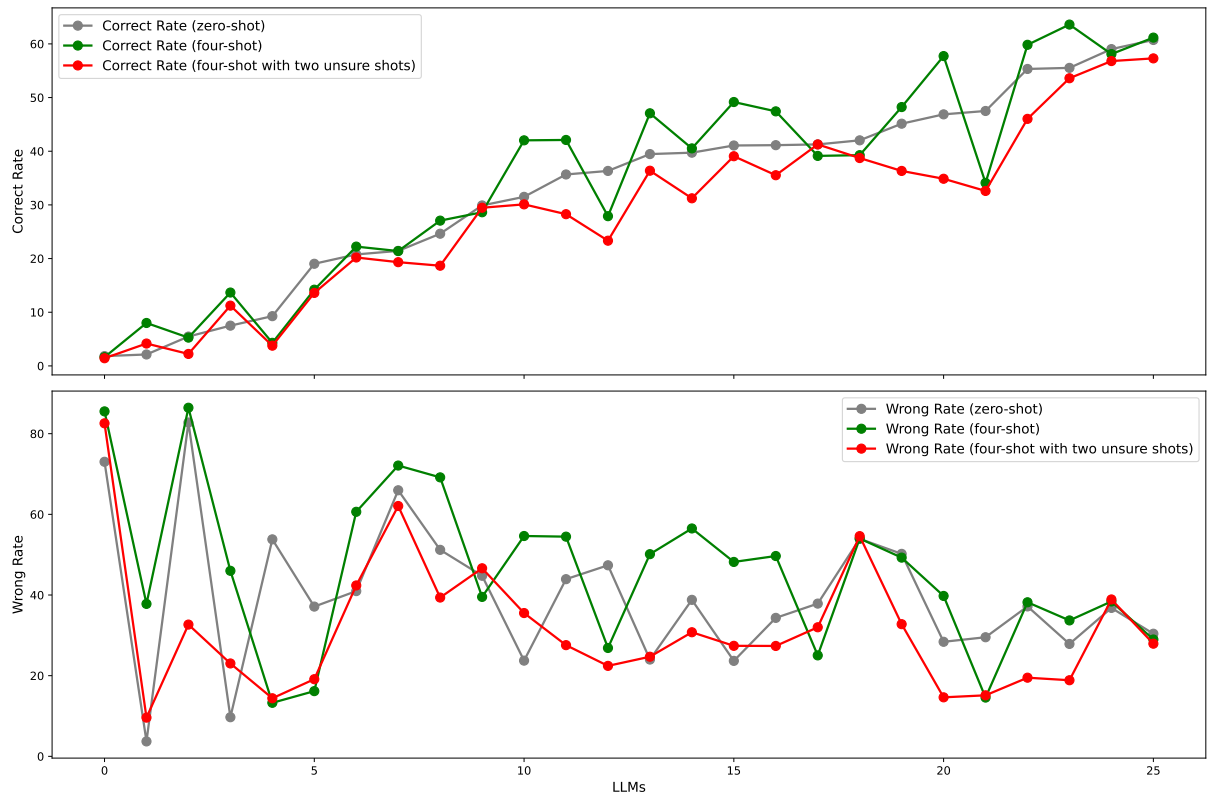


Figure 12: Correct Rate and Wrong Rate under different prompt settings for LLMs ranked in Correct Rate under zero-shot.